

```

library(dplyr)
library(janitor)
library(sf)
library(tmap)

filter <- dplyr::filter
rename <- dplyr::rename
select <- dplyr::select
summarise <- dplyr::summarise
`%nin%` = Negate(`%in%`)

# setwd("/Users/alexander/Downloads/Data")
setwd("~/Library/Mobile Documents/com~apple~CloudDocs/TAMU/Teaching/2025:2026/2026
Spring/DAEN ISEN 489/Assignment3_sol")

# FLOODING
# https://floodmaps.fema.gov/NFHL/status.shtml
floodplain <- st_read("./48041C_20250122/S_FLD_HAZ_AR.shp")

floodplain <-
  floodplain %>% filter(!st_is_empty(.)) %>%
  st_make_valid() %>%
  clean_names() %>%
  filter(st_geometry_type(.) %in% c("POLYGON", "MULTIPOLYGON"))

floodplain %>% st_drop_geometry() %>%
  count( fld_zone, zone_subty
  )
#   fld_zone          zone_subty    n
# 1      A          <NA>      138
# 2      AE      FLOODWAY      47
# 3      AE          <NA>     812
# 4      X 0.2 PCT ANNUAL CHANCE FLOOD HAZARD 1756
# 5      X      1 PCT FUTURE CONDITIONS      15
# 6      X      AREA OF MINIMAL FLOOD HAZARD  139

floodplain_class <-
  floodplain %>% mutate(
    fld_class = case_when(fld_zone %in% c("VE", "V", "AE", "AH", "A0", "A", "A99") ~
      "100Yr",
                        fld_zone == "X" & zone_subty == "1 PCT FUTURE CONDITIONS" ~
      "100Yr",
                        fld_zone == "X" & zone_subty == "0.2 PCT ANNUAL CHANCE FLOOD
HAZARD" ~ "500Yr",
                        TRUE ~ "Out")
  ) %>% filter(fld_class != "Out") %>%
  select(fld_zone, zone_subty, fld_class) %>%
  group_by(fld_class) %>%
  summarise() %>%
  st_make_valid()

# DON'T DO, SUPER SLOW
# tmap(floodplain_class) +
#   tm_polygons("fld_class")

# SUPPOSEDLY "FASTER", BUT STILL VERY SLOW
# qtm(floodplain_class, fill = "fld_class", style = "cobalt", crs = "+proj=eck4")

```

```
# st_write(floodplain_class, "floodplain_48041C_20250122.shp", delete_dsn = TRUE)

floodplain_class_simp <-
  smoothr::smooth(floodplain_class,
    method = "spline")

# st_write(floodplain_class_simp, "floodplain_class_simp_48041C_20250122.shp",
delete_dsn = TRUE)

as.numeric(object.size(floodplain_class)) /
  as.numeric(object.size(floodplain_class_simp))

# PARCELS
# https://brazoscad.org/tax-information/gis/

parcels <-
st_read("./Public_Parcel_Boundary_certified_2025/Public_Parcel_Boundary_certified.shp")

parcels_point <- parcels %>%
  st_make_valid() %>%
  clean_names() %>%
  select(state_cd, yr_built, state_cd, market, land_val, imprv_val) %>%
  st_point_on_surface()

parcels_point$state_cd_2 <- substr(parcels_point$state_cd, 1, 1)

table(parcels_point$state_cd_2, useNA = "always") %>% addmargins()
#      A      B      C      D      E      F      J      M  <NA>  Sum
# 55222 3270  6125  2408  4743  3702   49    3   130 75652

# ANALYSIS
parcels_point_floodplain_join <-
  st_join(parcels_point,
    st_transform(floodplain_class_simp, st_crs(parcels_point)))

table(parcels_point_floodplain_join$fld_class,
  useNA = "always") %>% addmargins()
# 100Yr 500Yr  <NA>  Sum
#  2486   456 72710 75652

with(
  parcels_point_floodplain_join,
  table(fld_class, state_cd_2, useNA = "always")
)

#      state_cd_2
# fld_class      A      B      C      D      E      F      J      M  <NA>
# 100Yr      847    73    442    520    483    109    4     0     8
# 500Yr      340    21     56     1     9     26    0     0     3
# <NA>     54035  3176  5627  1887  4251  3567   45     3   119

table(
  parcels_point_floodplain_join$fld_class,
  parcels_point_floodplain_join$state_cd_2,
  useNA = "always"
) %>% addmargins()

#      A      B      C      D      E      F      J      M  <NA>  Sum
```

# 100Yr	847	73	442	520	483	109	4	0	8	2486
# 500Yr	340	21	56	1	9	26	0	0	3	456
# <NA>	54035	3176	5627	1887	4251	3567	45	3	119	72710
# Sum	55222	3270	6125	2408	4743	3702	49	3	130	75652

```
# st_write(parcel_point_floodplain_join, "parcel_point_floodplain_join.shp",
delete_dsn = TRUE)
```

```
# save.image("~/Library/Mobile
Documents/com~apple~CloudDocs/TAMU/Teaching/2025:2026/2026 Spring/DAEN ISEN
489/Assignment3_sol/Assignment3_workspace.RData")
```

```
# ...
```

```
# ALTERNATIVELY ...
```

```
floodplain_100yr <- floodplain_class_simp |>
  filter(fld_class == "100Yr") |>
  st_transform(st_crs(parcel_point))
```

```
parcel_A <- parcel_point |>
  filter(state_cd_2 == "A")
```

```
# DON'T DO, VERY SLOW
```

```
# sf_use_s2(FALSE)
```

```
# As_and_100yr = st_intersection(parcel_A, floodplain_100yr)
```

```
# POINT-POLYGON
```

```
# st_intersects is typically used for point-in-polygon
```

```
# st_intersects(parcel_A, floodplain_100yr)
```

```
# st_within returns TRUE for points inside, FALSE for points on the boundary
```

```
# st_within(parcel_A, floodplain_100yr)
```

```
# NOT POINT-POLYGON
```

```
# st_touches will be FALSE
```

```
# st_touches(parcel_A, floodplain_100yr)
```

```
# ...
```

```
# st_overlaps(parcel_A, floodplain_100yr)
```

```
# st_contains(parcel_A, floodplain_100yr)
```

```
vector <- lengths(st_within(parcel_A, floodplain_100yr)) > 0
```

```
table(vector)
```

```
parcel_touching <- parcel_A[vector, ]
```

```
# OR
```

```
# As_and_100yr = st_intersection(filter(parcel_point, state_cd_2 == "A"),
#                               st_transform(filter(floodplain_class_simp, fld_class
# == "100Yr"), st_crs(parcel_point)))
```